

```
1  /*
2     यदा यदा हि धर्मस्य क्लान्भवति भारत।
3     अभ्युत्थानमधर्मस्य तदात्मानं सृजाम्यहम्॥
4
5     */
6     /*
7     ॐ त्र्यम्बकं यजामहे सुगन्धिं पुष्टिवर्धनम् ।
8     उर्वारुकमिव बन्धनान्मृत्योर्मुक्षीय माऽमृतात् ॥
9
10    */
11
12    #include <bits/stdc++.h>
13    using namespace std;
14
15    int partitionArray(vector<int> &v1, int lo, int hi) {
16        int pivot = v1[hi];
17        int i = lo - 1;
18        for (int j = lo; j < hi; j++) {
19            if (v1[j] < pivot) {
20                i++;
21                swap(a: [&] v1[i], b: [&] v1[j]);
22            }
23        }
24        swap(a: [&] v1[i + 1], b: [&] v1[hi]);
25        return i + 1;
26    }
27
28    void quickSort(vector<int> &v1, int lo, int hi) {
29        if (lo < hi) {
30            int pivotIndex = partitionArray(&v1, lo, hi);
31            quickSort(&v1, lo, pivotIndex - 1);
32            quickSort(&v1, pivotIndex + 1, hi);
33        }
34    }
35
36    int main() {
37        int n;
38        cout << "Enter size of vector: ";
39        cin >> n;
40        vector<int> v1(n);
41        cout << "Enter vector elements: ";
42        for (int i = 0; i < n; i++) {
43            cin >> v1[i];
44        }
45        cout << "Original array:" << endl;
46        for (int i = 0; i < n; i++) {
47            cout << v1[i] << " ";
48        }
49        cout << endl;
50        quickSort(v1, 0, n - 1);
51        cout << "Sorted array:" << endl;
52        for (int i = 0; i < n; i++) {
53            cout << v1[i] << " ";
54        }
55        cout << endl;
56        return 0;
57    }
```

D:\My_Programs\GitHub_Repository\CPP\CodeChef\Lab_3.1

Enter size of vector:5

Enter vector elements:10 8 93 3 2

Original array:
10 8 93 3 2

Sorted array:
2 3 8 10 93

Process finished with exit code 0

```
CodeChef - main
parallel_algorithms.md CodeGround_A.cpp Lab_3.cpp Lab_2.cpp Task_1.cpp Difficulty_1400_01.cpp Difficulty_1400_02.cpp
int main() {
    int n;
    cout << "Enter size of vector: ";
    cin >> n;
    vector<int> v1(n);
    cout << "Enter vector elements: ";
    for (int i = 0; i < n; i++) {
        cin >> v1[i];
    }
    cout << "Original array:" << endl;
    for (int el : v1) {
        cout << el << " ";
    }
    cout << endl;
    quickSort(&v1, 0, v1.size() - 1);
    cout << "Sorted array:" << endl;
    for (int el : v1) {
        cout << el << " ";
    }
    cout << endl;
    return 0;
}

D:\My_Programs\GitHub_Repository\CPP\CodeChef\Lab_3.cpp
Enter size of vector:5
Enter vector elements:10 8 93 3 2
Original array:
10 8 93 3 2
Sorted array:
2 3 8 10 93
Process finished with exit code 0
```